MATHEMATICS

SECTION – I
Single Correct Choice Type

This section contains 6 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. A bag contains 10 white and 10 black balls. A person draws two balls at a time without replacement and repeats the process till bag is empty. The probability that he draws the balls of same colour each time is

(a) $\left(\frac{2}{3}\right)^{10}$ (b) $\frac{{}^{10}C_5}{{}^{20}C_{10}}$

(c) $\left(\frac{2^{10}-1}{{}^{20}C_{10}}\right)$ (d) none of these

2. If $f(x)=\frac{\{x\}\ g(x)}{\{x\}\ g(x)}$ is a periodic function with period $\frac{1}{4}$, where $g(x)$ is a differentiable function, then (where {.} represents fractional part of x)

(a) $g'(x)=0$ has exactly three roots in $\left(\frac{1}{4},\frac{5}{4}\right)$

(b) $g(x)=0$ at $x=\frac{k}{4}$, where $k\in I$

(c) $g(x)$ must be non-zero function

(d) $g(x)$ must be periodic function

3. If $f'\left(\frac{x}{y}\right).f\left(\frac{y}{x}\right)=\frac{x^2+y^2}{xy}\ \forall\ x,y\in R^+$ and $f(1)=1$, then $f^2(x)$ is

(a) $x+\ln x$ (b) $x^2+2\ln x$

(c) $x^2+\ln x$ (d) $x+2\ln x$

4. If $(x^2+2x+4)^n=\sum_{r=0}^{2n}a_r x^r$, then $\sum_{r=n}^{2n}\left(\frac{a_{2n-r}}{a_r}\right)$ is

(a) $2^{2n+1}-1$ (b) $\frac{4^{n+1}-1}{3}$

(c) 4^{n-1} (d) $\frac{4^{n-1}+1}{2}$

5. The straight line $\frac{x}{4}+\frac{y}{3}=1$ intersects the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ at two points *A* and *B*, there is a point *P* on this ellipse such that the area of ΔPAB is equal to $6(\sqrt{2}-1)$, the number of such points (*P*) is

(a) 1 (b) 2

(c) 3 (d) 4

6. The number of ordered pair (*a*, *b*) where $a, b \in R$ such that $(a+ib)^{2010}=a-ib$ is

(a) 2010 (b) 2011

(c) 2009 (d) 2012

SECTION – II
Multiple Correct Choice Type

This section contains 5 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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7. Let $f(x)=|x|$, $g(n,x)=\sin\left(\pi\left([n]+[n]^2\right)^{1/x}\right)$ $h(x)=\frac{\ln\left(\frac{e^x+e^{2x}+e^{3x}}{3}\right)}{x}$, then

(where [.] represent greatest integer function)

(a) $\lim_{n\to\infty}\lim_{x\to 0} f(g(n,h(x)))=1$ (b) $\lim_{n\to\infty}\lim_{x\to 0} f(g(n,h(x)))$ does not exist

(c) $\lim_{n\to\infty}\lim_{x\to 0} g(n,h(x))$ does not exist (d) $\lim_{n\to\infty}\lim_{x\to 0} g(n,h(x))=1$

8. The curve $y=(|x|-1)\operatorname{sgn}(x-1)$ divides $\frac{9x^2}{64}+\frac{4}{25}y^2=\frac{1}{\pi}$ in two parts having area A_1 and A_2 (where $A_1 < A_2$), then

(a) $\frac{A_1}{A_2}=\frac{7}{13}$ (b) $\frac{A_1}{A_2}=\frac{3}{7}$

(c) $A_1=\frac{7}{3}$ (d) $A_2=\frac{13}{7}$

9. Let $f(n)$ be the sum of first *n* terms of sequence 0, 1, 1, 2, 2, 3, 3, 4, 4 ……., then

(a) $f(n)=\frac{n^2}{4}$, where *n* is an even number

(b) $f(n)=\frac{n^2-1}{4}$, where *n* is an odd number

(c) $f(n+m)-f(n-m)=nm$, where $n, m \in I^+ (n > m)$

(d) $f(n+m)-f(n-m)=\frac{4nm+1}{1}$, where $n, m \in I^+ (n > m)$

10. Let $f_1(x)=\sin^{-1}\left(\cos(\sin^2 x)\right)$, $f_2(x)=\cos^{-1}(\sin(\cos^2 x))$
$f_3(x)=\sin^{-1}\left(\cos(\cos^2 x)\right)$, $f_4(x)=\cos^{-1}(\sin(\sin^2 x))$, then choose the correct option(s)

(a) $\sum_{i=1}^{4} f_i(x)=2\pi-2$

(b) $\sum_{i=1}^{4}(-1)^i f_i(x)=0$

(c) $\sum_{i=1}^{4}(-1)^{\left[\frac{i}{3}\right]} f_i(x)=0$ (where [.] is greater integer function)

(d) $f_1(x) > f_2(x) \Rightarrow \cos 2x > 0$

11. If $f(x) = ax^2 + bx + c = 0$ have real roots and its coefficients are odd positive integers then

(a) $f(x) = 0$ always have irrational roots

(b) $\left| f\left(\frac{p}{q}\right) \right| \geq \frac{1}{q^2}$ where $p, q \in I$

(c) if $a.c = 1$, then equation must have exactly one root α such that $[\alpha] = -1$, where [.] is greatest integer function

(d) equation have rational roots

SECTION – III
Reasoning Type

This section contains 5 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

12. **Statement-1 :** $f(x) = \frac{\ln x}{(e^{-x} + 1)} (x > 1)$ is an increasing function.

Statement-2 : For $f(x) = \frac{g(x)}{h(x)}$, if $g(x)$ is an increasing function and $h(x)$ is a decreasing function, $g(x)$ and $h(x) > 0$, then $f(x)$ is an increasing function.

(a) A (b) B (c) C (d) D

13. **Statement-1 :** The minimum length of tangent drawn from point P on $3x+4y-20=0$ to the circle $x^2+y^2=1$ is $\sqrt{15}$.

Statement-2 : If tangent from point P touches the circle at T and secant passing through P intersect it at Q and R, then $PT^2 = PQ.PR$.

(a) A (b) B (c) C (d) D

14. **Statement-1 :** Let $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ and A_i is cofactor corresponding to a_i, then

$$\sum_{i=1}^{3} a_{3n+i} A_{3m+i} = 0, \quad (n \neq m \text{ and } n, m \in \{0, 1, 2\})$$

Statement-2 : If a determinant has two rows identical, then its value is zero.

(a) A (b) B (c) C (d) D

15. **Statement-1 :** $\tan\left(\frac{\pi}{4}\left(\frac{1+\sin^2 x}{1+\sin^2 y}\right)\right) + \tan\left(\frac{\pi}{4}\left(\frac{1+\cos^2 x}{1+\cos^2 y}\right)\right) > 1$ for $x, y \in \left(0, \frac{\pi}{2}\right)$.

Statement-2: If $f(x,y) = \left(\frac{1+\sin^2 x}{1+\sin^2 y} - 1\right)$, then $f(x,y).f\left(\frac{\pi}{2}-x, \frac{\pi}{2}-y\right) \leq 0 \ \forall \ x, y \in \left(0, \frac{\pi}{2}\right)$.

(a) A (b) B (c) C (d) D

16. Consider the sequence $\frac{{}^nC_0}{1.2}, \frac{{}^nC_1}{2.3}, \frac{{}^nC_2}{3.4}, \ldots\ldots\ldots$

Statement-1 : For $n = 12$, 6th term of sequence is greatest.

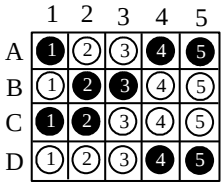
Statement-2: The general term of sequence is $T_r = \frac{{}^nC_{r-1}}{r.(r+1)}$ (for $r = 1, 2, \ldots\ldots, n$).

(a) A (b) B (c) C (d) D

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1.

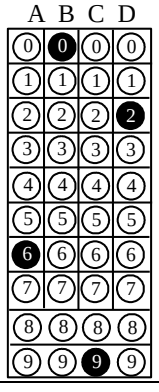
Column-I		Column-II
(A)	Let $S_n = \cos^4\theta + \cos^4\left(\theta + \frac{2\pi}{n}\right) + \cos^4\left(\theta + \frac{4\pi}{n}\right) + \ldots$ upto n terms, then S_n can be	**1.** 2
(B)	If in a ΔABC $\cos mA + \cos mB + \cos mC = 1 + 4\sin\frac{mA}{2}\sin\frac{mB}{2}\sin\frac{mC}{2}$, then m can be	**2.** 3
(C)	If $\cos^n x - \sin^n x = 1$ have 11 roots in $\left[0, \frac{23\pi}{2}\right)$, then n can be	**3.** 1
(D)	The values of m for which equation $(5m - m^2)^2 \sin^2 x - 10\sin x(5m - m^2) + 24 = 0$, has exactly three solutions in $[0, 2\pi]$ are	**4.** 6 **5.** 5

2.

Column-I	**Column-II**
(A) $\lim\limits_{n\to\infty}\dfrac{\left(\sum\limits_{x=1}^{n}x^4\right)\left(\sum\limits_{x=1}^{n}x^5\right)}{\left(\sum\limits_{x=1}^{n}x^t\right)\left(\sum\limits_{x=1}^{n}x^{9-t}\right)}=\dfrac{4}{5}$, then t can be	**1.** 6
(B) If m be the slope of tangent to the curve $x^y=y^x$ at (e, e), then $(3-m)$ can be	**2.** 2
(C) If $f:R-\{0\}\to R,\ f(x)f(y)=f(xy)+3\left(\dfrac{1}{x}+\dfrac{1}{y}\right)$, then $2\left\lvert f\left(-\dfrac{1}{2}\right)\right\rvert$ can be	**3.** 3
(D) $\lim\limits_{n\to\infty}\left\{(\sqrt{2}+1)^{n!+K}\right\}=1$, then K can be (where $\{.\}$ is a fractional part function)	**4.** 4 **5.** 7

SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



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1. If the line $3x+4y=12$ intersects the hyperbola at P & P' and its asymptotes at Q and Q' then find the value of $\frac{PQ}{P'Q'}+\frac{PQ'}{P'Q}$.

Space for work

mathongo

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2. Three students appear in three examinations. If the number of possible ways in which neither any student fails in all examinations nor there is an examination, which is not cleared by any student is n, then find $\frac{n}{53}$.

Space for work

mathongo

3. Let $f(x) = 30 - 2x - x^3$, then find the number of positive integral values of x which satisfies $f(f(f(x))) > f(f(-x))$.

Space for work

mathongo

4. Let $A=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$, if $\det.(A+A^2+A^3+A^4+\ldots+A^n)=64$, then find the value of n.

Space for work

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